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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IRB: The Shanghai Changhai Hospital Ethics Committee approved the studies , and we obtained written informed consent from all participants .
Consent: The Shanghai Changhai Hospital Ethics Committee approved the studies , and we obtained written informed consent from all participants .
Inclusion and Exclusion Criteria
not detected.
Attrition
Additional 24 participants (12 patients with TNBC and 12 healthy donors) donated serum, and 16 participants (eight patients with TNBC and eight health donors) donated serum exosomes these were used for data validation.
Sex as a biological variable
not detected.
Subject Demographics
Age: Patient Serum Samples Serum samples from patients with TNBC (n = 30 aged 36-88 years median age, 52) were obtained from College of Basic Medical Sciences, Navy Medical University (supplemental Table
Randomization
The peptides in each individual sample and the pooled samples were randomly distributed and labeled across three 10-plex TMT experiments.
Blinding
not detected.
Power Analysis
not detected.
Replication

Statistical Analysis All functional experiments were repeated independently at least three times.
Number: Statistical Analysis All functional experiments were repeated independently at least three times.
Cell Line Authentication
Contamination: All cells were cultured in a humidified incubator with 5 CO2 at 37 degC and confirmed to be free of mycoplasma contamination .

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Next , the membranes were blocked with 5 nonfat dry milk in Tris-buffered saline containing 0.1 Tween-20 (TBST) at room temperature for 1 h and subsequently incubated overnight at 4 degC with primary antibodies against TSG101 (sc-7964 Santa Cruz Biotechnology) , HSP70 (4873S CST) , Alix (2171S CST) , CD63 (ab59479 Abcam) , Calreticulin (12238T CST)	TSG101		
	HSP70		
	CD63		
	Calreticulin		
Immunogold Labeling and Electron Microscopy The grids were placed into blocking buffer for 1 h and then into a solution of the primary antibody at the appropriate dilution overnight at 4 degC (anti-CD151 , 96283S	anti-CD151		
Experimental Models: Cell Lines			
Cell Lines and Cell Culture Triple-negative human metastatic breast carcinoma cells (BT549 , MDA-MB-468 , and MDA-MB-231) and estrogen receptor expression breast carcinoma cells (T47D and MCF7) were purchased from American Type Culture Collection .	MDA-MB-468		Suggestion: RRID:CVCL_0419(link)
	MCF7		Suggestion: RRID:CVCL_0031(link)
MCF-7 , MDA-MB-468 , and MDA-MB-231 cells were maintained in Dulbeccos modified Eagles medium with the addition of 10 (vv) fetal bovine serum (FBS) , 100 mgml streptomycin , and 100 Uml penicillin .	MCF-7		Suggestion: RRID:CVCL_0031(link)
T47D cells were grown in RPMI1640 supplemented with 10 FBS , 10 mgml insulin , 100 Uml penicillin , and 100 mgml streptomycin .	T47D		Suggestion: RRID:CVCL_0553(link)
BT549 cells were grown in RPMI1640 supplemented with 10 FBS , 100 Uml penicillin , and 100 mgml streptomycin .	BT549		Suggestion: RRID:CVCL_1092(link)
Migration and Invasion Assays For migration assay, 2 x 105 MDA-MB-468 and 1 x 105 MDA-MB-231 cells were plated in transwell chambers (6-well transwell chambers, 8-mm pore size 3428 Corning).	MDA-MB-231		Suggestion: RRID:CVCL_0062(link)

Recombinant DNA			
Guide RNAs (5-TGAGTGGATCCGCTCACAGG or 5-GCTGGTAGTAGGCGTAGGCG) targeting a sequence within the exon of the CD151 gene were cloned into lentiCRISPRv2 (plasmid no. 52961)	lentiCRISPRv2		
Software and Algorithms			
Tandem MS data were obtained using an Orbitrap Fusion Lumos instrument, and proteins were identified using Proteome Discoverer 2.2.	Proteome Discoverer		Suggestion: Proteome Discoverer, RRID:SCR_014477(link)
Western Blotting Samples containing 15 mg of exosomal proteins derived from the serum or cell lysates of a study subject were denatured with SDS (Sigma) loading buffer , boiled for up to 5 min , separated by SDS-PAGE , and transferred onto a polyvinylidene fluoride membrane (Bio-Rad) at 30 V for 60 min .	Sigma		Suggestion: sigma-tau (RRID:SCR_000488)(link)
For each of the TMT experiments , raw files from the fractions were merged and searched with the SEQUEST HT search engine with a Homo sapiens Swiss-Prot UniProt protein database downloaded on February 2019 (95,556 entries) .	UniProt		
The false discovery rate (FDR) was calculated using Percolator algorithm provided by PD .	Percolator		
Bioinformatics Analysis The screening of differentially expressed proteins (DEPs) was carried out by using the statistical analysis tool of Metaboanalyst (https:www.metaboanalyst.ca).	Metaboanalyst		
	https:www.metaboanalyst.ca		
For Gene Ontology analysis , Metascape was used to convert UniProtKB accessions into Entrez Gene IDs (18) , and Entrez Gene IDs were mapped to Gene Ontology terms for cellular components by FunRich software (version 3.1.3) (19 , 20) .	Entrez Gene		
Volcano plots and Venn diagrams were drawn using R package ggplot2	ggplot2		Suggestion: ggplot2 (RRID:SCR_014601)(link)
(version 3.2.1) (21) and R package VennDiagram (version 1.6.20) (22)	VennDiagram		
Heat maps were generated using hierarchical clustering analysis based on Euclidean distance and R package pheatmap (version 1.0.12) (23) .	pheatmap		
Enrichment of functions and signaling pathways of the target proteins	Metascape		

was determined using Metascape (http:metasape.org) (18) and the enrichKEGG function in the R package clusterProfiler (version 3.12.0) (24) .	clusterProfiler		Suggestion: clusterProfiler (RRID:SCR_016884)(link)
The STRING database (http:string-db . org) was applied to identify protein networks , which were visualized by the plugin cytoHubba (version 0.1) (25) in Cytoscape software (version 3.8.2 Cytoscape Consortium) (26)	STRING		
	Cytoscape		Suggestion: Cytoscape (RRID:SCR_003032)(link)
Representative fields were photographed, and the number of cells per field was counted by ImageJ software (U.S. National Institutes of Health).	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
Data were analyzed using GraphPad Prism 8 software (GraphPad Software, Inc) and shown as mean +- SEM.	GraphPad Prism		
	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
DEPs were determined by t test and further characterized through gene enrichment and network analysis.	t test
Significant p value was obtained by Students t test (patients with TNBC versus health donors)	Students t test
Multiple sets of data were analyzed by one-way or two-way ANOVA, and unpaired Students t test was used for analyses of two sets of data.	ANOVA

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